HENSELIAN PAIRS ARE PRESERVED UNDER FILTERED COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Henselian pair, defined through the lifting of factorisations of monic polynomials modulo the ideal. definition used in the same section, source lines 2860-2883, immediately before the counted statement. Supplies the property that has to be preserved and the form in which the lifting is verified.
- An ideal lies in the Jacobson radical if and only if every element of $1 + I$ is a unit. the author cites Algebra, Lemma algebra-lemma-contained-in-radical (tag 0AME) at proof line 2897. Turns the invertibility statement proved for $1 + I$ into the radical condition.
- Filtered colimits: an element of a filtered colimit is the image of an element of some stage, and the same for a finite system of elements and relations. used at proof lines 2893-2910 to descend f, the factorisation and the Bezout relation to one index j. Is the mechanism that reduces the verification to a single henselian pair.

Source reference. Original lemma 0FWT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19858–19858. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13402–13402. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *The property of being Henselian is preserved under filtered colimits of pairs. More precisely, let J be a directed set and let (A_j, I_j) be a system of henselian pairs over J. Then $A = \operatorname{colim} A_j$ equipped with the ideal $I = \operatorname{colim} I_j$ is a henselian pair (A, I).*

Proof. If $u \in 1+I$ then for some $j \in J$ we see that u is the image of some $u_j \in 1+I_j$. Then u_j is invertible in A_j by Algebra, Lemma 0AME and the assumption that I_j is contained in the Jacobson radical of A_j. Hence u is invertible in A. Thus I is contained in the Jacobson radical of A (by the lemma).

Let $f \in A[T]$ be a monic polynomial and let $\overline{f} = g_0 h_0$ be a factorization with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$. Write $1 = g_0 g_0' + h_0 h_0'$ for some $g_0', h_0' \in A/I[T]$. Since $A = \operatorname{colim} A_j$ and $A/I = \operatorname{colim} A_j/I_j$ are filtered colimits we can find a $j \in J$ and $f_j \in A_j$ and a factorization $\overline{f}_j = g_{j,0} h_{j,0}$ with $g_{j,0}, h_{j,0} \in A_j/I_j[T]$ monic and $1 = g_{j,0} g_{j,0}' + h_{j,0} h_{j,0}'$ for some $g_{j,0}', h_{j,0}' \in A_j/I_j[T]$ with $f_j, g_{j,0}, h_{j,0}, g_{j,0}', h_{j,0}'$ mapping to f, g_0, h_0, g_0', h_0'. Since (A_j, I_j) is a henselian pair, we can lift $\overline{f}_j = g_{j,0} h_{j,0}$ to a factorization over A_j and taking the image in A we obtain a corresponding factorization in A. Hence (A, I) is henselian. $\square$